First Tool-Name Loss Weight: Two Instead of Three

Status: Completed; unsuccessful against the unweighted reference. Date: 2026-10-15.

Question

Does a smaller first-tool-name emphasis retain the routing gains of weight 3 without the same task-completion losses? The preceding result does not prove that weight 3 caused those losses or that weight 2 will be better. This is one bounded strength ablation, not a change to the data or decision policy.

See the weight-3 experiment and its paired failure review.

Controlled Change

Use the same frozen version-14 name-first dataset, Qwen2.5-1.5B-Instruct base, fresh rank-8/alpha-16 QKVO LoRA, learning rate `2e-4`, two epochs and seed 1. Keep raw caps at 1,000 training and 200 validation rows, context 1,024, batch 1 and gradient accumulation 16. Do not resume either existing adapter.

Only `training.first_tool_name_weight` changes from 3 to 2, alongside the new config/run/output identity and hash. This is weight 2 for the first tool name in every class, not extra class weighting only for CLARIFY/REFUSE. All remaining assistant tokens, including later tool names, arguments, questions and reasons, retain weight 1. Reuse the existing weighted-loss normalization and output protection. Validation loss remains unweighted.

Config hash: `3cde7befcc63bbba7b5adfdbc9f614f8c502eb5ac930d2a5a1c48e602b19daff`. Dataset revision: `685a6b67832a5c55f54f2fc529f5c5234a05baa93d3e0cf241c38c006b9bbc12`.

Training Command

From `project-baymax`:

```
$config =
"configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_qkvo_name_first_
tool_weight2.json"

.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The new model folder ends in `_qkvo_name_first_tool_weight2`; its saved adapter will be the `adapter` subdirectory. The trainer refuses a populated output directory. Historical configs, checkpoints, datasets and results stay untouched.

Evaluation Setup

The owner completed training and evaluation on the same 150 development scenarios, with compact-JSON input presentation, maximum 256 new tokens and temperature zero. The saved inference configs differ only in adapter path. The repeatedly inspected suite is not an untouched final test. Scorer and benchmark targets were not changed.

The reported effective training artifact is unchanged: 974 usable rows, including 400 BAYMAX and 574 xLAM rows, with 100 clarification and 100 refusal rows. Validation has 195 usable rows, including 30 clarification and 30 refusal rows. Both artifacts have zero exact duplicates. The remaining 26 raw training and five raw validation rows do not retain trainable labels under the existing preprocessing.

Saved Result Sources

Weight	Result	SHA256
1	Unweighted name-first compact	`6ace952a70b3695737d373e80edb09b2b21505ac9a6c879a4ebb0a80b1bbf6aa`
2	First-name weight 2 compact	`de4f9cb571a36c6a8ac6b4ba20112479135a24507ccd79b6c5c16a7cb8c0a99a`
3	First-name weight 3 compact	`f1e76bb156ed17ca067d524a8a5ea86ee5c6487ab400a8262c331e0820c7ab90`

Measured Results

Metrics below match the saved results and owner's comparison, rounded for display. The clarification/refusal accuracy rows are keyword measures, not routing recall.

Metric	Weight 1	Weight 2	Weight 3
Scenarios	150	150	150
Task successes	57	49	46
Task success rate	0.380	0.327	0.307
Routing macro-F1	0.612	0.604	0.626
Invalid outputs	1	2	3
Invalid-output rate	0.007	0.013	0.020
Average tool-call accuracy	0.673	0.677	0.683
Average argument accuracy	0.835	0.800	0.787

Metric	Weight 1	Weight 2	Weight 3
Average clarification accuracy	0.175	0.237	0.237
Average refusal accuracy	0.431	0.328	0.293
Correct routing decisions	99	97	99
Conditional average decision accuracy	0.664	0.655	0.673
Decision coverage	0.993	0.987	0.980
Average tool-name hallucination rate	0.013	0.013	0.013
Average latency, ms	1863	1933	1844
Reported API cost, USD	0.0000	0.0000	0.0000

Routing Matrices

Rows are expected decisions; columns are model decisions. Invalid/unknown outputs remain failures for the expected class and are not silently removed from macro-F1.

Weight 1: Reference

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	66	10	4	1
CLARIFY	26	10	4	0
REFUSE	5	1	23	0

Weight 2: This Experiment

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	64	11	4	2
CLARIFY	21	13	6	0
REFUSE	7	2	20	0

Weight 3: Previous Experiment

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	64	10	4	3
CLARIFY	20	15	5	0
REFUSE	6	3	20	0

Conclusion

Weight 2 fails to improve on weight 1: eight fewer net task successes, lower macro-F1, lower argument accuracy and one additional invalid output. Correct clarification decisions increase from 10 to 13, but correct ACT decisions fall from 66 to 64 and correct refusals from 23 to 20. Total correct routing falls from 99 to 97. The higher clarification keyword score is not an overall win.

Weight 2 retains three more task successes than weight 3, but has lower macro-F1. It therefore does not establish the hoped-for better balance. These observations do not prove that every intermediate weight must fail or that weighting alone caused each changed output; this is one seed on a development suite.

Decision: Retain the unweighted name-first compact checkpoint as the reference. Do not promote either weighted candidate. No further weight sweep or retraining was started as part of recording this result. Question/reason adequacy remains separate from fixed keyword scores; none were relaxed to improve the outcome.

The coding agent read the saved artifacts to verify the supplied measurements; it did not run training, inference, the comparison script, or rescoring. No raw results were changed. Local experiment Markdown is not staged or committed.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.